Pre-Calculus 11 - Quadratic Functions Practice Test v1

Rubric								
Outcome	0	1	2	3	4	5	6	7
Understanding and Solving								
Mathematical Communication								

St. George's School - Academic Integrity

Academic Dishonesty includes any conduct with the intent to gain an unfair advantage in connection with an academic assessment. Academic Dishonesty can occur in many ways. Common forms of Academic Dishonesty include, but are not limited to, cheating, falsification, plagiarism, and tampering. Cheating occurs when an individual undermines the integrity of an assessment (including homework and other assignments, reports, projects, quizzes, tests, exams, or other forms of performance evaluations).

Examples of cheating include:

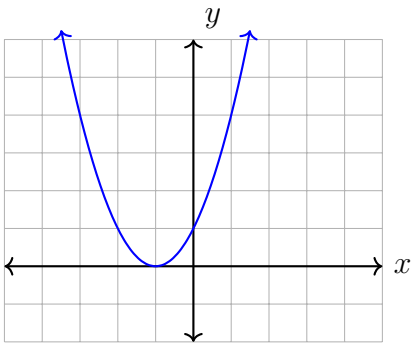
- Copying any task of an assessment;
- Allowing others to copy any task of an assessment;
- Improperly giving or receiving assessment information;
- Using unauthorized resources for or during an assessment;
- Skipping classes to avoid an assessment

Name : _____

Teacher's name: _____

Level 1 to Level 2 Questions

1. Given the graph of the quadratic function below, identify the following:
   (a) The coordinates of the vertex.
   (b) The equation of the axis of symmetry.
   (c) The domain and range.

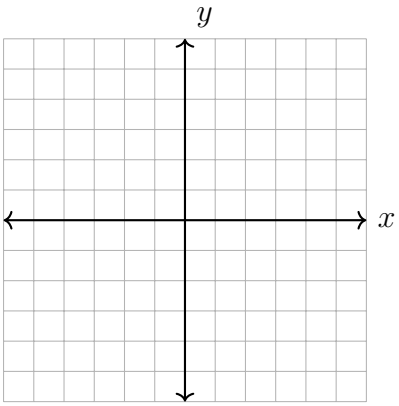


2. State the vertex, axis of symmetry, and direction of opening for the following functions.

   a) $y = -2(x-3)^2 + 4$

   b) $y = \frac{1}{2}(x+1)^2 - 5$

   c) Sketch both parts a) and b) on the same graph to the right.



3. Solve the following equations using the square root method. Leave answers in exact form.

   a) $(x-4)^2 = 25$

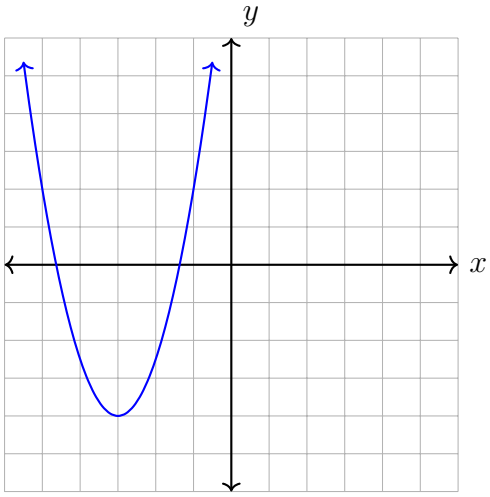
   b) $2(x+3)^2 - 10 = 0$

Level 3 to Level 4 Questions

4. Describe the transformations required to change the graph of $y = x^2$ into the graph of $y = -3(x + 2)^2 - 5$. Use proper mathematical language.

5. Determine the equation of the quadratic function in vertex form given the following properties: Vertex at $(-3, 0)$ and y-intercept at $(0, -9)$.

6. Write the equation for the given parabola in vertex form.



7. Convert $y = x^2 - 10x + 19$ to Vertex Form by **completing the square**.

8. Find the vertex of $y = -2x^2 - 8x + 5$ using any method.

9. Use the discriminant to determine the nature of the roots for $2x^2 - 3x + 5 = 0$ (do not solve).

10. Find the value(s) of k such that the equation $x^2 + kx + 9 = 0$ has exactly **one real root**.

Level 5 to Level 6 Questions

11. Solve the following quadratic equation. Write answers in exact form.

$$\frac{1}{3}(2x-3)^2 = 8$$

12. A concert venue has 3,000 seats and usually sells 2000 tickets when the ticket price is \$120. Market research indicates that for every \$10 increase in ticket price, 100 less tickets are sold. Determine the ticket price that will maximize revenue and how many tickets will be sold at that price.

13. A farmer has 250 meters of fencing to enclose a rectangular field. She also wants to add a fence dividing the field in half parallel to one side. Determine the dimensions that maximize the total area.

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Level 7 Question(s)

14. Given a, b, c are real numbers and $a \neq 0$. For what values of a, b, c will the equation

$$a(x-b)^2 = c$$

a) have two distinct real answers

b) have no real answers

c) have one real answer (or two identical real answers)